

A Formal Dynamic Schema Evolution Model for Hypermedia Databases

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Abstract

A hypermedia database schema is a meta-information which defines the structures of applications and objects. A dynamic schema evolution model is an abstract machine which describes changes of schemata. Unfortunately, because of the lack of formal data structures of hypermedia databases, it is very difficult to describe the dynamic changes of a hypermedia schema. The research presented in this paper proposes a formal approach to define the model of dynamic schema information. That is, a complete formal structure of a hypermedia database on the basis of previous research[WH93, WH95, Wan96, Wan97], and the schema operations defined for the schema evolution. The significant features of this research are, firstly, a formalized model to define the dynamic schema structure, secondly, a theoretical approach to manage the global schema of a distributed system, and finally, a design strategy to integrate the Web databases to support a wide range of application areas.

Keyword Formal Data Model, Multimedia, Dynamic Schema, Distributed Databases and Object Orientation.

1 Introduction

When a hypermedia application involves a large amounts of different types of multimedia information, it becomes more and more difficult to maintain the complex object types due to the lack of a powerful mechanism to define and maintain the structure of nodes and links. Consequently, a well-known problem in hypermedia, *the disorientation of hypermedia navigation*, appeared[Hal88, Hal91]. The essence of the disorientation is that the multimedia information types are unstructurally organized. The solution to this problem relies on the way of how to store information and how to describe relationships among nodes. Databases can be naturally chosen as the most suitable candidates for supporting complex hypermedia applications. This is because databases themselves are well structured for modelling a complex application structures. Since 90's, the author has focused on the data structure of linking databases with hypermedia to support complex application environments[HW92, Wan93, WH93, WH95]. Several enhanced hypermedia data models are developed[Wan96, Wan97]. In this paper, a further study of the dynamic hypermedia schema based on an extended hypermedia model[WH95] is discussed. In particular, we will illustrate the mechanisms of how to maintain the semantics of hypermedia application schemata in order to keep the consistency the node and link structures.

This paper is organized as follows. Section 2 reviews the semantic constructors of an extended hypermedia data model in order to illustrate the hypermedia schema structure which is described in section 3. Section 4 describes some related works and we conclude our discussion in section 5.

2 A Hypermedia Model

The dynamic hypermedia schema evolution model is set up on the basis of an existing extended hypermedia data model which is an object oriented data model capturing both semantics of hypermedia and object databases.¹ There three basic constructors in this model. They are *node constructor*, *composite node constructor* and *schema constructor*. These constructors are used as essential semantic building blocks for constructing a hypermedia application structure on the basis of object oriented databases. In the following review of this model, we will focus on the semantic aspects of these constructors and show how these constructors can formally present a hypermedia application.²

2.1 Relationships

The main problem of the current hypermedia systems to model an application structure is that they lack a powerful facility to reveal relationships among node. Thus, our extended hypermedia model focus on this aspect and reveal how the node structure can easily capture those semantic relationships among nodes.

In the conventional hypermedia approach, components of an application are stored in nodes and the nodes are connected by links. Because links are used simply to represent a fact that there exists a connection among nodes. It is very difficult for both designers and users to know the exact semantic relationship between nodes. Thus, it depends heavily on the human beings abilities to reveal relevant information behind nodes and links. It is therefore useful if a hypermedia node can automatically represent some semantic meanings of application structures.

There are two major relationship types between hypermedia components. They are part-of and inheritance relationships. In the extended hypermedia data model we developed, we define two node constructors to reveal such internal structure of a node.

Component Node Constructor

This is the first node constructor which reveals the part-of relationship between nodes. In general, a conventional hypermedia node is constructed by multimedia information. The multimedia data types within a node can be used as a either the destination or target of links. Thus, a node structure can be described by figure 1.

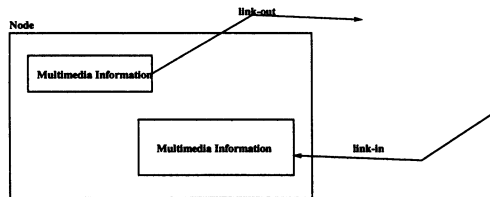


Figure 1: A conventional node structure

In the conventional approach, we cannot find out the exact semantic relationship between the node and its multimedia components. The node is a virtual object. It is only used as a concept to describe an information container. However, if we impose the part-of relationship between components and its container, we can modify the above picture into a more semantic node type as described in figure 2.

¹This is a unified data model called HyperDB which was developed by the author in 1997[Wan97].

²A formal specification language Z is used to describe these constructors. Z is the formal specification language developed by the Oxford University in the late seventies. Now, it is widely used both in industry and academia.